

DUNCO HMS

Hospital Management System

COMPREHENSIVE IMPLEMENTATION PROPOSAL

A complete, Kenya-ready hospital management solution with

M-Pesa Payments • SHA/SHIF Integration • DHA Connectivity • AI Assistants

Prepared for: [Hospital Name]

Prepared by: Dunco Web Solutions

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1. Executive Summary

Dunco HMS is a comprehensive, web-based Hospital Management System designed to centralize and streamline all hospital operations — from patient registration and clinical care to billing, pharmacy, laboratory, HR, and management reporting.

Built for Kenya. Ready for Africa.

Dunco HMS is purpose-built for the Kenyan healthcare environment with deep integration into M-Pesa, SHA/SHIF, DHA, and AI-powered clinical support.

Kenya-Specific Integrations:

- **M-Pesa (Safaricom Daraja API)** — Accept payments via STK Push. Patients pay consultation fees, pharmacy bills, and lab charges directly from their phone. Real-time callbacks and automatic invoice reconciliation.
- **SHA / SHIF (Social Health Authority)** — Full eligibility verification, member lookup, authorization requests, claims submission, and coverage tracking through the DHA Health Interoperability Gateway.
- **DHA (Digital Health Agency)** — Patient verification against the national Client Registry, facility lookup, provider verification, and clinical document exchange via the Digital Health Superhighway.
- **AI-Powered Healthcare** — Built-in AI assistants (Elliana D Virtual Nurse) for patient queries, appointment suggestions, diagnosis support, and predictive analytics — available in English and Swahili.

Core Capabilities:

- Centralized patient management across OPD, IPD, and emergency
- Integrated billing with M-Pesa, insurance, cash, and card support
- Pharmacy and inventory control with expiry tracking and stock management
- Laboratory and radiology with test requests, results, and PDF reports
- Human resources management including payroll, attendance, leave, and appraisals
- Insurance and claims management with SHA integration
- Telemedicine via Zoom integration
- Patient self-service portal for appointments, results, and billing
- Marketing suite with social media, SEO, AI content writer, and campaigns
- Comprehensive reporting with custom report builder and BI dashboard

All packages include FREE domain registration (.or.ke or .co.ke) for Year 1 and UNLIMITED professional email accounts (e.g., info@hospital.or.ke, doctor@hospital.or.ke, pharmacy@hospital.or.ke) — no per-user charges, no hidden email costs.

2. About Dunco HMS

Dunco HMS is developed by **Dunco Web Solutions**, a Kenyan software company specializing in healthcare technology. The system is purpose-built for the Kenyan market with deep integration into local payment systems, insurance frameworks, and health data infrastructure.

Technology

Component	Technology
Backend	Laravel 12 (PHP 8.2+)
Frontend	Blade, Alpine.js, Tailwind CSS
Database	MySQL 8.0+ with Redis
Authentication	Laravel Breeze + Sanctum
Authorization	Spatie Permission (21 roles, 95+ permissions)
PDF	DomPDF
Excel	Maatwebsite Excel
Backups	Spatie Laravel Backup

Scale

Metric	Count
Controllers	153
Models	131
Blade Views	457
Database Migrations	163
Configurable Modules	75+
Granular Permissions	95+
User Roles	21

3. Kenya-Specific Integrations

Dunco HMS is the only hospital management system built from the ground up for the Kenyan healthcare environment. No other HMS in the market offers this combination of M-Pesa payments, SHA/SHIF insurance, DHA national connectivity, and AI-powered clinical support.

M-Pesa Payment Integration (Safaricom Daraja API)

- **STK Push** — Patients receive a payment prompt on their phone and enter their M-Pesa PIN
- **Real-Time Callbacks** — System receives instant confirmation from Safaricom
- **Automatic Reconciliation** — Payments automatically linked to invoices
- **Transaction Tracking** — Full M-Pesa transaction history visible in the system
- **Multi-Fee Support** — Consultation fees, lab tests, pharmacy, admissions, deposits
- **Coverage Checks** — Verify SHA coverage before processing patient payments

SHA / SHIF Integration (Social Health Authority)

- **Patient Eligibility Verification** — Real-time check via DHA Health Interoperability Gateway
- **Member Lookup** — Find patients by National ID, CR ID, or SHA number
- **Authorization Requests** — Submit pre-authorization for procedures
- **Claims Submission** — Submit claims directly to SHA for reimbursement
- **Coverage Tracking** — Monitor claim status, approvals, and payments
- **Service Codes** — Complete Kenya SHA procedure and service code database
- **ICD-10 Codes** — Standard diagnostic codes for claims and reporting

DHA Integration (Digital Health Agency)

- **National Client Registry** — Verify patients against Kenya's national health database
- **Facility Registry** — Look up facility details and FRN codes
- **Provider Registry** — Verify healthcare provider credentials
- **Document Interchange** — Send and receive clinical documents (HL7/FHIR)
- **Biometric Verification** — Fingerprint and facial recognition against national database
- **Emergency Access** — Emergency access tokens for urgent patient care

AI-Powered Healthcare

- **Elliana D Virtual Nurse** — 24/7 AI chatbot handling patient queries, appointment booking, and medical information in English and Swahili
- **AI Appointment Suggestions** — Intelligent scheduling based on doctor availability and department load
- **AI Diagnosis Support** — Symptom-based condition matching to assist clinicians
- **AI Predictive Analytics** — Bed occupancy forecasting and revenue projections
- **AI Content Writer** — Generates hospital marketing content and social media posts

4. Modules & Capabilities

Clinical Operations

Module	Description
Patient Registration	Full CRUD, search, DHA verification, auto-invoicing
OPD / IPD	Outpatient and inpatient management with bed assignment
Doctors & Scheduling	Profiles, departments, schedules, availability
Nurses	Duty roster, ward assignments, departments
Appointments	Walk-in + online booking with M-Pesa/SHA payment
Queue Management	Token generation, display boards, kiosk mode
Prescriptions	E-prescriptions, templates, PDF, dispensing
Medical History	Vitals, allergies, drug interactions
Diagnosis	ICD-10 codes, categories, patient diagnoses
Discharge Summary	PDF generation, workflow management
Birth / Death Reports	Certificates, statistical reports
Operation Theatre	Rooms, scheduling, instruments, sterilization
Blood Bank	Donors, inventory, requests, stock levels
Ambulance & Emergency	Fleet management, call tracking, emergency admissions
Bed Management	Types, assignments, visualization, IoT sensors
Consent / Vaccination / Mortuary / CSSD / MRD	Complete clinical support modules

Diagnostics

Module	Description
Laboratory	Tests, requests, results, technicians, PDF reports
Radiology	Tests, requests, results, PDF reports
Investigation Reports	Cross-module investigation tracking

Pharmacy & Inventory

Module	Description
Medicines	Catalog, categories, brands, stock tracking
Prescription Dispensing	Stock deduction, billing integration
Drug Interactions	Real-time interaction checking
Inventory Management	Multi-store, stock movements, transfers

Suppliers & Purchase Orders	Supplier management, approval workflow
Expiry Tracking	Automated expiry alerts

Billing & Finance

Module	Description
M-Pesa Payments	STK Push, callbacks, reconciliation
Invoices & Receipts	Line items, PDF, thermal printing
Payments	Cash, card, M-Pesa, insurance
Chart of Accounts	Ledger, trial balance, P&L, balance sheet
Expenses & Income	Categories, reports, tracking
Multi-Currency	Live exchange rates, conversion
Insurance Claims	Submit, approve, reject, SHA integration

Insurance & SHA

Module	Description
Insurance Providers	Company management, coverage settings
Patient Insurance	Policy tracking, eligibility
Claims Management	Full claims lifecycle with PDF
SHA Member Management	Registration, verification, authorization
SHA Service Codes	Kenya SHA procedure codes
ICD-10 Database	Diagnostic codes for claims

Human Resources

Module	Description
Employees	Profiles, photo upload, user account creation
Departments & Designations	Organization structure
Attendance	Daily tracking, reports
Leave Management	Types, requests, approval workflow
Payroll	Salary generation, reports
Shifts & Roster	Shift types, roster builder
Appraisals	Performance reviews
Training	Programs, enrollments, certificates
Recruitment	Job postings, applications
HR Reports	8+ report types with PDF export

AI Features

Feature	Description
Elliana D Virtual Nurse	AI chatbot for patient queries (English + Swahili)
AI Appointment Suggestions	Intelligent scheduling optimization
AI Diagnosis Support	Symptom-based condition matching
AI Predictive Analytics	Bed occupancy and revenue forecasting
AI Content Writer	Marketing content generation

Additional Modules

Module	Description
Telemedicine	Video consultations via Zoom
Patient Portal	Self-service booking, results, billing
Marketing Suite	CMS, blog, social media, campaigns, SEO
RFID / IoT	Tag management, bed monitoring sensors
Biometric Security	Fingerprint/face enrollment and verification
ID Cards	Patient and employee cards with QR codes
Custom Report Builder	Create, schedule, and generate custom reports
BI Dashboard	Interactive charts and analytics
Multi-Branch	Manage multiple hospital locations
Module Manager	Enable/disable features per department

5. Pricing Packages

All three tiers include ALL modules — clinical, pharmacy, lab, radiology, billing, HR, insurance/SHA, AI, telemedicine, CMS, marketing, IoT/RFID, reporting, and analytics. No features are locked or restricted in any tier.

Tier 1: Clinic

1-3 departments, <20 staff

KES 300,000

One-time implementation

Annual renewal: KES 120,000/year

- All modules included
- Installation & setup
- Configuration & branding
- Data migration
- User training
- FREE domain (.or.ke)
- UNLIMITED email accounts

Tier 2: Hospital

4-10 departments, 20-100 staff

KES 550,000

One-time implementation

Annual renewal: KES 180,000/year

- All modules included
- Extended implementation
- Comprehensive training
- Priority support
- FREE domain (.or.ke)
- UNLIMITED email accounts

Tier 3: Enterprise

10+ departments, 100+ staff, multi-branch

KES 800,000

One-time implementation

Annual renewal: KES 300,000/year

- All modules included

- Enterprise implementation
- Full training program
- Premium SLA (2-hr response)
- FREE domain (.or.ke)
- UNLIMITED email accounts

Total First-Year Investment

Package	One-Time	Annual	Total Year 1
Tier 1 (Clinic)	KES 300,000	KES 120,000	KES 420,000
Tier 2 (Hospital)	KES 550,000	KES 180,000	KES 730,000
Tier 3 (Enterprise)	KES 800,000	KES 300,000	KES 1,100,000

6. Domain & Professional Email

FREE Domain Registration (Year 1)

Item	Details
Domain Options	.or.ke (recommended for organisations), .co.ke (commercial), .ac.ke (academic), .go.ke (government)
Year 1 Cost	FREE — registered and managed by Dunco Web Solutions
Year 2+ Cost	KES 2,000/year (renewed annually)
DNS Management	Included
SSL Certificate	FREE (Let's Encrypt, auto-renewed)

UNLIMITED Professional Email

Feature	Details
Email Accounts	UNLIMITED — create as many as you need
Domain	yourhospital.or.ke (or .co.ke)
Examples	info@hospital.or.ke, doctor@hospital.or.ke, pharmacy@hospital.or.ke, reception@hospital.or.ke, billing@hospital.or.ke, lab@hospital.or.ke
Webmail	Yes — access from any browser
Mobile Access	Yes — Android, iPhone, Outlook, Thunderbird
Storage	10 GB per account
IMAP/SMTP	Yes — works with any email client
Calendar	Included
Admin Panel	Create, delete, manage accounts
Spam Protection	Included
Annual Cost	FREE — hosted on your included server, no per-user charges

Your hospital email addresses will look professional:

admin@hospital.or.ke • reception@hospital.or.ke • dr.mwangi@hospital.or.ke • pharmacy@hospital.or.ke • billing@hospital.or.ke • lab@hospital.or.ke • info@hospital.or.ke

7. Infrastructure & Hosting

Component	Tier 1	Tier 2	Tier 3
CPU	2 vCPU	4 vCPU	8 vCPU
RAM	4 GB	8 GB	16 GB
Storage	80 GB SSD	160 GB NVMe	320 GB NVMe
Database	MySQL 8.0	MySQL 8.0 + Redis	MySQL 8.0 + Redis
Bandwidth	Unmetered	Unmetered	Unmetered
Backup Storage	30 GB	100 GB	500 GB
Monthly Hosting	KES 4,000	KES 6,000	KES 10,000

Why healthcare needs reliable hosting: Patient data availability is critical. Downtime affects patient care. Data loss is unacceptable. Security breaches have legal consequences. Performance affects user adoption.

8. Implementation Timeline

Phase	Duration	Milestone
Infrastructure Setup	Week 1	Server ready, domain registered
Application Deployment	Weeks 1-2	System accessible online
Third-Party Integration	Weeks 2-3	M-Pesa, SMS, SHA/DHA live
Hospital Configuration	Weeks 3-4	Branding, departments, services loaded
Data Migration	Weeks 4-5	Historical data imported and verified
Training	Weeks 5-6	All user groups trained
Testing / UAT	Weeks 6-7	UAT sign-off
Go-Live	Week 7	System operational
Post-Go-Live Support	Weeks 8-12	Stabilization and optimization

Total: 12 weeks from contract signing to full stabilization

9. User Training

User Group	Sessions	Duration	Key Topics
System Administrators	4	3 hrs each	System config, users, roles, backups, troubleshooting
IT Staff	3	3 hrs each	Server management, deployment, monitoring
Reception / Front Desk	3	2 hrs each	Patient registration, appointments, queue management
Doctors	2	2 hrs each	Patient records, prescriptions, scheduling, telemedicine
Nurses	2	2 hrs each	Patient care, vitals, ward management, duty roster
Pharmacy	3	2 hrs each	Medicine management, dispensing, inventory, prescriptions
Laboratory	3	2 hrs each	Test requests, results entry, reports
Radiology	2	2 hrs each	Test requests, results, reports
Billing / Cashiers	3	2 hrs each	Invoices, payments, M-Pesa, receipts
Finance / Accounts	3	2 hrs each	Chart of accounts, ledger, P&L, expenses
HR	3	2 hrs each	Employees, attendance, leave, payroll
Insurance / SHA Desk	2	2 hrs each	Claims, SHA verification, authorization
Management	2	2 hrs each	Dashboards, reports, analytics, KPIs
Marketing / CMS	2	2 hrs each	Website, blog, social media, campaigns

Total: 39 training sessions across 14 user groups

10. Security & Data Protection

Implemented Security Measures

- Password hashing (bcrypt, 12 rounds)
- Role-based access control (21 roles, 95+ permissions)
- CSRF protection on all forms
- XSS protection via Blade auto-escaping
- SQL injection prevention via Eloquent ORM
- Session management with regeneration on login
- Audit logging of all critical actions
- Module-level access control
- Biometric data encryption (AES-256-CBC)
- Secure file upload handling

Data Protection

- Daily automated backups with 30-day retention
- Database and file backups included
- Restore capability on demand
- Audit trail on all patient data access
- Role-based data visibility controls

11. Support & Maintenance

Feature	Basic KES 58,000/yr	Standard KES 82,000/yr	Premium KES 120,000/yr
Response Time	24 hours	8 hours	2 hours
Support Hours	Business hours	Extended hours	24/7
Remote Support	Email only	Email + phone	Email + phone + remote
On-Site Support	—	—	Monthly visits
Emergency Support	—	—	24/7 included
Bug Fixes	Yes	Yes	Yes (priority)
Security Updates	Yes	Yes	Yes
Application Updates	Yes	Yes	Yes
Database Maintenance	—	Yes	Yes
Performance Monitoring	—	Yes	Yes
Quarterly Review	—	Yes	Yes

12. Payment Terms

Tier 1 (Clinic) — KES 300,000

Milestone	%	Amount	Due
Contract Signing	50%	KES 150,000	Day 0
UAT Sign-off	30%	KES 90,000	Week 7
Go-Live + 30 days	20%	KES 60,000	Week 11

Tier 2 (Hospital) — KES 550,000

Milestone	%	Amount	Due
Contract Signing	40%	KES 220,000	Day 0
Infrastructure Ready	20%	KES 110,000	Week 2
UAT Sign-off	25%	KES 137,500	Week 7
Go-Live + 30 days	15%	KES 82,500	Week 11

Tier 3 (Enterprise) — KES 800,000

Milestone	%	Amount	Due
Contract Signing	35%	KES 280,000	Day 0
Infrastructure Ready	20%	KES 160,000	Week 2
UAT Sign-off	25%	KES 200,000	Week 7
Go-Live + 30 days	20%	KES 160,000	Week 11

13. Licensing & Ownership

- **Software License:** Perpetual, non-exclusive, non-transferable license to use Dunco HMS at the licensed facility
- **Source Code:** Remains property of Dunco Web Solutions. NOT transferred to client
- **Customizations:** Custom work performed for the client becomes part of the licensed system
- **Data:** All hospital data remains property of the hospital
- **Intellectual Property:** Dunco Web Solutions retains all IP rights
- **Restrictions:** Client may not redistribute, resell, or sublicense the software

Warranty

- 30-day warranty from go-live date
- Covers bugs and defects in delivered functionality
- Does not cover client modifications or third-party outages

14. Client Responsibilities

1. Provide hospital data (departments, services, fees, staff list) within 2 weeks of contract signing
2. Obtain and provide M-Pesa production credentials from Safaricom
3. Obtain and provide SHA/DHA facility credentials from Kenya Ministry of Health
4. Designate a project coordinator
5. Ensure stakeholder availability for training
6. Provide feedback during UAT within 5 business days
7. Ensure reliable internet connectivity (minimum 10 Mbps)
8. Provide computers/tablets for system access
9. Make payment milestones on time

Exclusions

- Third-party subscription fees (M-Pesa transaction fees, SMS charges, AI API usage, Zoom)
- Hardware procurement (computers, tablets, printers)
- Network infrastructure (LAN, WiFi, cabling)
- RFID hardware and IoT sensors
- Biometric hardware
- Legal and regulatory compliance consultation

15. Acceptance & Sign-off

Item	Client Sign-off	Date
Infrastructure Setup		
Application Deployment		
Third-Party Integration		
Hospital Configuration		
Data Migration		
Training Completion		
UAT Sign-off		
Go-Live Approval		

Authorized Client Representative	Dunco Web Solutions
Name	Name
Title	Title
Signature	Signature
Date	Date